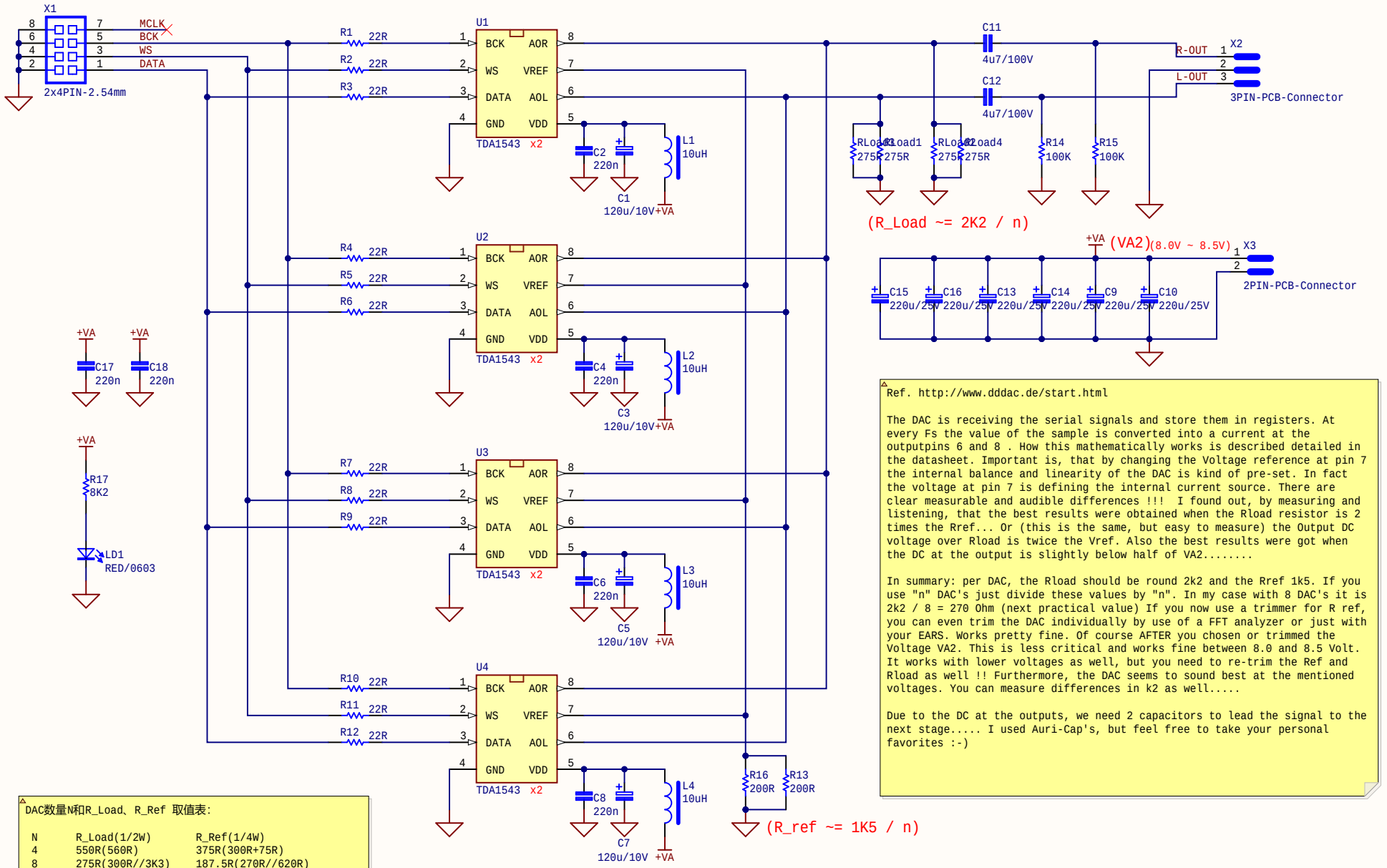


Ref. http://www.dddac.de/start.html

The DAC is receiving the serial signals and store them in registers. At every Fs the value of the sample is converted into a current at the outputpins 6 and 8 . How this mathematically works is described detailed in the datasheet. Important is, that by changing the Voltage reference at pin 7 the internal balance and linearity of the DAC is kind of pre-set. In fact the voltage at pin 7 is defining the internal current source. There are clear measurable and audible differences !!! I found out, by measuring and listening, that the best results were obtained when the Rload resistor is 2 times the Rref... Or (this is the same, but easy to measure) the Output DC voltage over Rload is twice the Vref. Also the best results were got when the DC at the output is slightly below half of VA2.....

In summary: per DAC, the Rload should be round 2k2 and the Rref 1k5. If you use "n" DAC's just divide these values by "n". In my case with 8 DAC's it is 2k2 / 8 = 270 Ohm (next practical value) If you now use a trimmer for R ref, you can even trim the DAC individually by use of a FFT analyzer or just with your EARS. Works pretty fine. Of course AFTER you chosen or trimmed the Voltage VA2. This is less critical and works fine between 8.0 and 8.5 Volt. It works with lower voltages as well, but you need to re-trim the Ref and Rload as well !! Furthermore, the DAC seems to sound best at the mentioned voltages. You can measure differences in k2 as well.....

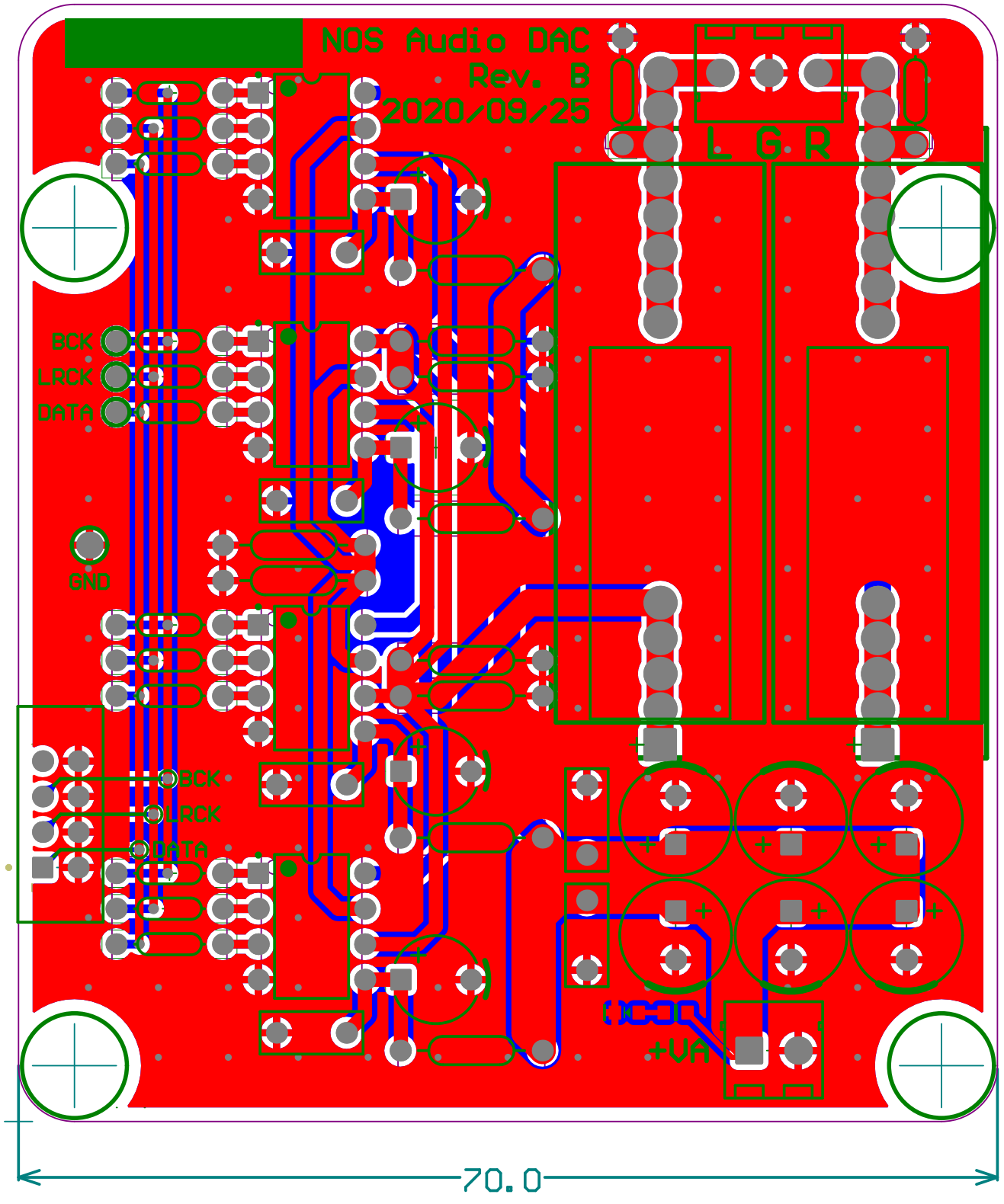
Due to the DC at the outputs, we need 2 capacitors to lead the signal to the next stage..... I used Auri-Cap's, but feel free to take your personal favorites :-)

DAC数量N和R_Load、R_Ref 取值表:

N	R_Load(1/2W)	R_Ref(1/4W)
4	550R(560R)	375R(300R+75R)
8	275R(300R//3K3)	187.5R(270R//620R)
12	183R(270R//560R)	125R(150R//750R)

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设计	Leon Lee	2020/06/14	华南理工大学 South China University of Technology	
审核				

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